

Sriguhan Ramesh
Machine Learning Engineer

Phone: +91 8277679780, +91 7397228147

Email: sriguhan@yandex.com, sriguhan@gmail.com

Career Objective

To secure a challenging position where I can effectively contribute my skills for the development of the organization while enhancing my existing skill set by collaborating with dedicated professionals.

Educational Qualifications

Qualification	Institution	CGPA
Masters in Manufacturing Engineering	Madras Institute of Technology (Anna University)	9.2
Bachelor of Engineering in Production Engineering	Madras Institute of Technology (Anna University)	7.53

Skills

- **Machine Learning Pipelines & MLOps:** Expertise in building and deploying ML training and inference pipelines using Vertex AI and Kubernetes.
- **LLM Inference Pipelines:** Proficient in setting up LLM inference servers with Faster Whisper, Text Generation Inference, and Llama CPP.
- **Deep Learning & Model Fine-Tuning:** Experience with PyTorch and Huggingface for tasks such as NLP (NER, REL) and time-series classification.
- **Containerization:** Developed Docker containers for deploying ML models, LLMs, and Falcon apps on Kubernetes.
- **Cloud Platforms:** Hands-on experience with Azure Cognitive Services (Speech to Text, Text to Speech, Translation) and Vertex AI; fair experience with AWS.
- **Model Deployment:** Skilled in deploying LLMs using RESTful API frameworks like FastAPI, BlackSheep, and Falcon, alongside web servers like Gunicorn and Hypercorn.

- **System Integration:** Integrated MongoDB and PostgreSQL into production workflows.
 - **RAG Systems:** Implemented retrieval-augmented generation (RAG) systems for conversational AI.
 - **CI/CD Pipelines:** Leveraged GitHub for CI/CD automation.
 - **Programming & Tools:** Proficient in Python, Pymongo, Bash scripting, and PySpark for data processing and analysis.
 - **Neural Network Frameworks:** Deep knowledge of PyTorch and Keras for developing custom models.
 - **Databases:** Extensive experience with MongoDB and fair familiarity with PostgreSQL.
-

Career History

- **Design Engineer (Automation)** - Suba Solutions Pvt Ltd (June 2017 – Nov 2020)
 - **Electrical Engineer** - Servo Hydraulics Pvt Ltd (Feb 2021 – Sep 2021)
 - **Co-Founder** - Botomaite Innovations Pvt Ltd (Oct 2021 – May 2022)
 - **Machine Learning Engineer** - NeuranceAI Technologies Pvt Ltd (June 2022 – Present)
-

Relevant Work Experience

- Fine-tuned Speech Recognition models using Huggingface.
 - Conducted NLP tasks like NER, REL, and assertion extraction on clinical texts using Spacy and HuggingFace.
 - Set up cloud VMs for deploying ML models.
 - Trained models like CatBoost and RocketClassifier for time-series classification.
 - Built training pipelines for text generation LLMs.
 - Deployed LLMs for production using RESTful API frameworks and web servers.
 - Configured MongoDB locally and as a cluster for API integrations.
-

Projects Undertaken

Project 1: Regional Language Speech Recognition

- Fine-tuned Wav2Vec models on proprietary datasets.
- Developed an API pipeline for seamless integration with an Android application.

Project 2: Medical Entity Recognition

- Fine-tuned SciSpacy NER models for medical entity recognition.
- Built pipelines to process clinical text and output structured data for database integration using ETL scripts.

Project 3: Mortality Prediction in ESKD Patients

- Designed a predictive model for mortality risk in End-Stage Kidney Disease (ESKD) patients using time-series analysis.
- Calculated a comprehensive health score based on dialysis routines and medical data.

Project 4: AI Medical Assistant for Chronic Conditions

- Developed an AI-powered medical assistant addressing patients' chronic health and nutritional queries.
- Integrated the assistant with WhatsApp using open-source LLMs and a RAG framework.

Project 5: Medical AI Coach and Assistant

- Created a Medical AI Coach for surgical patients.
 - Enabled multilingual interaction through text and voice.
-

Place: Bengaluru

Date: